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PROJECT REPORT

Work Done:

A two-terminal material-engineered semiconductor PN junction diode was simulated using Synopsys Sentaurus TCAD. Semiconductor regions across Silicon (Si), Germanium (Ge), Gallium Arsenide (GaAs), and 4H-Silicon Carbide (4H-SiC) were generated using Sentaurus Structure Editor (SDE). The effect of semiconductor energy bandgap, intrinsic carrier density, and carrier mobility was studied for forward turn-on knee voltage, reverse saturation leakage current, and avalanche breakdown. The I–V characteristics were obtained using Sentaurus Device (SDevice) and overlaid in Sentaurus Visual (SVisual).

SDE Code used for generating Silicon Semiconductor Structure:

```
(sde:clear)

;=====
; PN JUNCTION DIODE - MATERIAL-ENGINEERED STRUCTURE
;=====

; Silicon dimensions
(define Wdiode 1.0)
(define Hdiode 1.0)

; Silicon doping
(define Na 1e17)
(define Nd 1e16)

;=====
; SILICON REGIONS
;=====
(sdegeo:create-rectangle
(position 0.0 0.0 0.0)
(position 0.5 Wdiode 0.0)
"Silicon"
"R.P_Anode"
)
(sdegeo:create-rectangle
(position 0.5 0.0 0.0)
(position Hdiode Wdiode 0.0)
"Silicon"
"R.N_Cathode"
)

;=====
; CONTACTS
;=====

; Left side / Top edge = Anode contact
(sdegeo:set-contact
(find-edge-id
(position 0.0 0.5 0.0)
)
"Top"
)
```

```

; Right side / Bottom edge = Cathode contact
(sdegeo:set-contact
(find-edge-id
(position Hdiode 0.5 0.0)
)
"Bot"
)

;=====
; SILICON DOPING
;=====

(sdedr:define-constant-profile
"AnodeProf"
"BoronActiveConcentration"
Na
)
(sdedr:define-constant-profile-region
"AnodeDop"
"AnodeProf"
"R.P_Anode"
)

(sdedr:define-constant-profile
"CathodeProf"
"PhosphorusActiveConcentration"
Nd
)
(sdedr:define-constant-profile-region
"CathodeDop"
"CathodeProf"
"R.N_Cathode"
)

;=====
; MESH
;=====

(sdedr:define-refinement-size
"DiodeRef"
(/ Wdiode 20.0)
(/ Hdiode 20.0)
0.05
(/ Wdiode 40.0)
(/ Hdiode 40.0)
0.02
)
(sdedr:define-refinement-material
"DiodeRef"
"DiodeRef"
"Silicon"
)

;=====
; BUILD MESH
;=====

(sde:build-mesh "n@node@")

```

Why only Silicon is generated in SDE?

In this project, only the primary semiconductor structure is generated in Sentaurus Structure Editor (SDE). The purpose of SDE is to define the semiconductor geometry, doping profile, contacts and mesh required for the device simulation. The individual material parameters are not created as separate geometrical files because the main parameters being studied are the semiconductor bandgap, intrinsic carrier density, and carrier mobilities, rather than varying physical dimensions. Instead, the material properties are defined in Sentaurus Device (SDevice). The metallurgical junction boundary, contacts, and mesh remain identical, while the material model is assigned to Silicon, Germanium, Gallium Arsenide, and 4H-Silicon Carbide in SDevice. Therefore, the device physics and I–V characteristics can be directly compared under identical geometric conditions ($1.0\text{ }\mu\text{m} \times 1.0\text{ }\mu\text{m}$).

SDevice Code for Silicon:

```
#setdep @previous@
```

```
File {  
  Grid= "@tdr@"  
  Current= "@plot@"  
  Plot= "@tdrdat@"  
  Output= "@log@"  
}
```

```
Electrode {  
  {  
    Name = "Top"  
    Voltage = 0  
  }  
  {  
    Name = "Bot"  
    Voltage = 0  
  }  
}
```

```
Physics {  
  Mobility(  
    PhuMob  
    HighFieldSaturation(GradQuasiFermi)  
  )  
  Recombination (  
    SRH  
  )  
  EffectiveIntrinsicDensity (  
    BandGapNarrowing (oldSlotboom)  
  )  
  Fermi  
}
```

```
Plot {
TotalCurrent/Vector
eCurrent/Vector
hCurrent/Vector
ElectricField/Vector
Potential
SpaceCharge
eDensity
hDensity
eMobility
hMobility
DonorConcentration
AcceptorConcentration
ConductionBand
ValenceBand
eQuasiFermi
hQuasiFermi
}
```

```
Math {
Transient = BE
eMobilityAveraging = ElementEdge
hMobilityAveraging = ElementEdge
ParallelToInterfaceInBoundaryLayer(FullLayer -ExternalBoundary)
ComputeGradQuasiFermiAtContacts= UseQuasiFermi
WeightedVoronoiBox
AutoCNPMinStepFactor = 0
AutoNPMInStepFactor = 0
-PlotLoadable
SimStats
ExitOnFailure
Digits = 5
ErrRef(electron) = 1e8
ErrRef(hole) = 1e8
Iterations = 20
NotDamped = 100
RHSMIn = 1e-8
```

```

EquilibriumSolution(Iterations=100)
Extrapolate
RefDens_eGradQuasiFermi_ElectricField_HFS = 1e8
RefDens_hGradQuasiFermi_ElectricField_HFS = 1e8
Method = ParDiSo
NumberOfThreads = 4
ParallelLicense (Wait)
Wallclock
}

Solve {
Coupled (
Iterations = 100
LineSearchDamping = 1e-4
) {
Poisson
}
Coupled (Iterations = 100) {
Poisson
Electron
Hole
}
Quasistationary (
InitialStep = 1e-3
Increment = 1.41
MinStep = 1e-7
MaxStep = 0.05
Goal {
Name = "Top"
Voltage = -20.0
}
) {
Coupled {
Poisson
Electron
Hole
}
}
Quasistationary (
InitialStep = 1e-3
Increment = 1.41
MinStep = 1e-7
MaxStep = 0.05

```

```
Goal {  
  Name = "Top"  
  Voltage = 1.0  
}  
{  
  Coupled {  
    Poisson  
    Electron  
    Hole  
  }  
}
```

SDevice Code for Germanium:
#setdep @previous@

```
File {  
  Grid= "@tdr@"  
  Current= "@plot@"  
  Plot= "@tdrdat@"  
  Output= "@log@"  
}
```

```
Electrode {  
  {  
    Name = "Top"  
    Voltage = 0  
  }  
  {  
    Name = "Bot"  
    Voltage = 0  
  }  
}
```

```
Physics {  
  Material = "Germanium"  
  Mobility(  
    PhuMob  
    HighFieldSaturation(GradQuasiFermi)  
  )  
  Recombination (  
    SRH  
  )  
  EffectiveIntrinsicDensity (  
    BandGapNarrowing (oldSlotboom)  
  )  
  Fermi  
}
```

```
Plot {
TotalCurrent/Vector
eCurrent/Vector
hCurrent/Vector
ElectricField/Vector
Potential
SpaceCharge
eDensity
hDensity
eMobility
hMobility
DonorConcentration
AcceptorConcentration
ConductionBand
ValenceBand
eQuasiFermi
hQuasiFermi
}
```

```
Math {
Transient = BE
eMobilityAveraging = ElementEdge
hMobilityAveraging = ElementEdge
ParallelToInterfaceInBoundaryLayer(FullLayer -ExternalBoundary)
ComputeGradQuasiFermiAtContacts= UseQuasiFermi
WeightedVoronoiBox
AutoCNPMinStepFactor = 0
AutoNPMInStepFactor = 0
-PlotLoadable
SimStats
ExitOnFailure
Digits = 5
ErrRef(electron) = 1e8
ErrRef(hole) = 1e8
Iterations = 20
NotDamped = 100
RHSMIn = 1e-8
EquilibriumSolution(Iterations=100)
Extrapolate
```

```
RefDens_eGradQuasiFermi_ElectricField_HFS = 1e8
RefDens_hGradQuasiFermi_ElectricField_HFS = 1e8
Method = ParDiSo
NumberOfThreads = 4
ParallelLicense (Wait)
Wallclock
}
```

```
Solve {
Coupled (
Iterations = 100
LineSearchDamping = 1e-4
) {
Poisson
}
Coupled (Iterations = 100) {
Poisson
Electron
Hole
}
Quasistationary (
InitialStep = 1e-3
Increment = 1.41
MinStep = 1e-7
MaxStep = 0.05
Goal {
Name = "Top"
Voltage = -10.0
}
) {
Coupled {
Poisson
Electron
Hole
}
}
Quasistationary (
InitialStep = 1e-3
Increment = 1.41
MinStep = 1e-7
MaxStep = 0.05
Goal {
Name = "Top"
Voltage = 1.0
}
)
}
```



```
{
Coupled {
Poisson
Electron
Hole
}
}
}
```

SDevice Code for Gallium Arsenide & 4H-Silicon Carbide:

```
#setdep @previous@
```

```
File {
Grid= "@tdr@"
Current= "@plot@"
Plot= "@tdrdat@"
Output= "@log@"
}
```

```
Electrode {
{
Name = "Top"
Voltage = 0
}
{
Name = "Bot"
Voltage = 0
}
}
```

```
Physics {
Material = "GaAs"
Mobility(
PhuMob
HighFieldSaturation(GradQuasiFermi)
)
Recombination (
SRH
Radiative(B_rad = 7.2e-10)
)
EffectiveIntrinsicDensity (
BandGapNarrowing (oldSlotboom)
)
Fermi
}
```

```
Plot {
TotalCurrent/Vector
eCurrent/Vector
hCurrent/Vector
ElectricField/Vector
Potential
SpaceCharge
}
```

```

eDensity
hDensity
eMobility
hMobility
DonorConcentration
AcceptorConcentration
ConductionBand
ValenceBand
eQuasiFermi
hQuasiFermi
}

Math {
  Transient = BE
  eMobilityAveraging = ElementEdge
  hMobilityAveraging = ElementEdge
  ParallelToInterfaceInBoundaryLayer(FullLayer -ExternalBoundary)
  ComputeGradQuasiFermiAtContacts= UseQuasiFermi
  WeightedVoronoiBox
  AutoCNPMinStepFactor = 0
  AutoNPMInStepFactor = 0
  -PlotLoadable
  SimStats
  ExitOnFailure
  Digits = 5
  ErrRef(electron) = 1e8
  ErrRef(hole) = 1e8
  Iterations = 20
  NotDamped = 100
  RHSMIn = 1e-8
  EquilibriumSolution(Iterations=100)
  Extrapolate
  RefDens_eGradQuasiFermi_ElectricField_HFS = 1e8
  RefDens_hGradQuasiFermi_ElectricField_HFS = 1e8
  Method = ParDiSo
  NumberOfThreads = 4
  ParallelLicense (Wait)

```

```

Wallclock
}

Solve {
Coupled (
Iterations = 100
LineSearchDamping = 1e-4
) {
Poisson
}
Coupled (Iterations = 100) {
Poisson
Electron
Hole
}
Quasistationary (
InitialStep = 1e-3
Increment = 1.41
MinStep = 1e-7
MaxStep = 0.05
Goal {
Name = "Top"
Voltage = -50.0
}
) {
Coupled {
Poisson
Electron
Hole
}
}
Quasistationary (
InitialStep = 1e-3
Increment = 1.41
MinStep = 1e-7
MaxStep = 0.05
Goal {
Name = "Top"
Voltage = 1.5
}
) {
Coupled {
Poisson
Electron
Hole
}
}
}

```

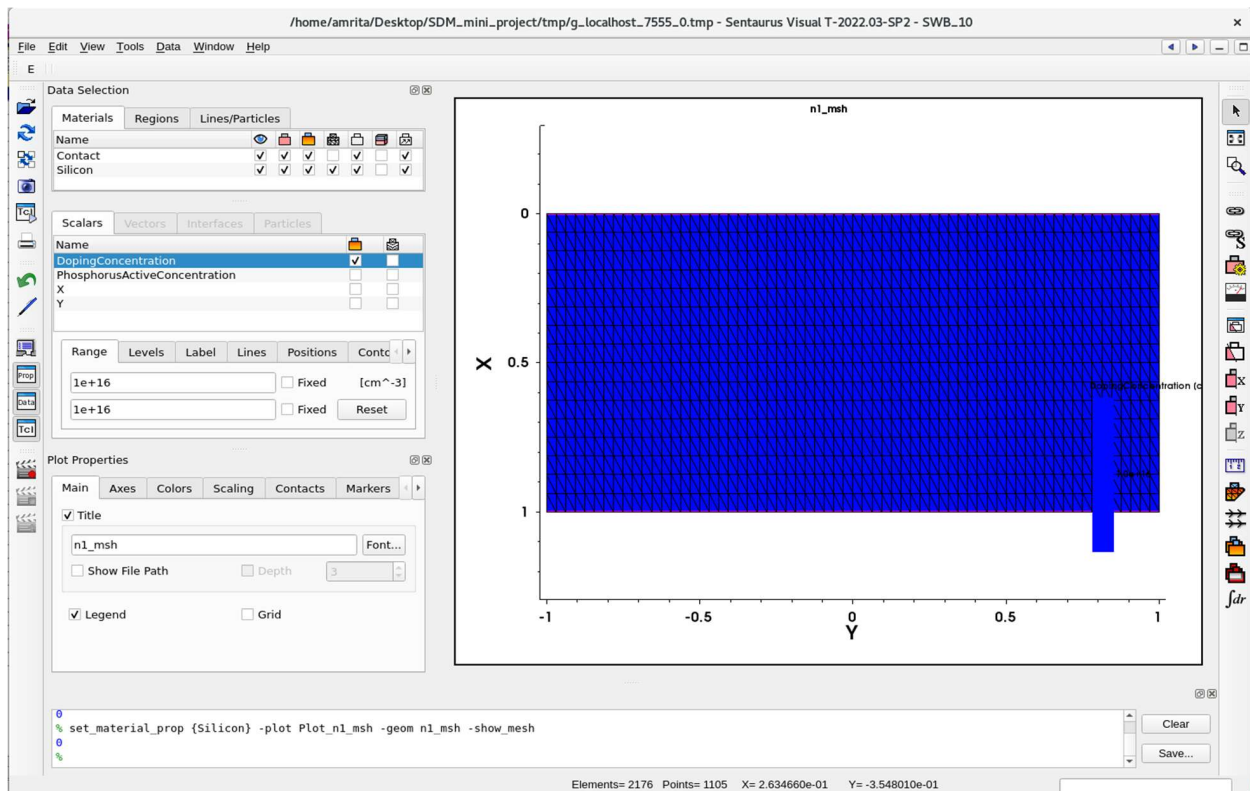
What is happening in the SDevice code ?

The SDevice code is used to simulate the electrical behaviour of the Silicon structure created in SDE. First, the mesh and other required files from SDE are loaded. The Top and Bottom sides of the Silicon are defined as contacts. The Top contact is made the Anode contact and the Bottom contact is made the Cathode. Four separate simulations are done for Germanium, Silicon, Gallium Arsenide, and 4H-Silicon Carbide under identical geometries.

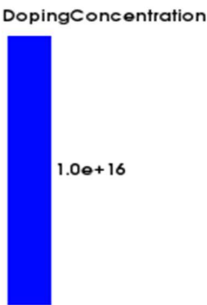
The Physics section tells Sentaurus which models to use for carrier movement and recombination. The Math section contains the settings needed to solve the simulation. In the Solve section, the device is first brought to its initial condition and then the voltage at the Top contact is varied from reverse bias to forward bias. The current produced by the device is recorded for each voltage. Finally, the voltage and current values are plotted in Sentaurus Visual to compare the I-V curves of Germanium, Silicon, Gallium Arsenide, and 4H-Silicon Carbide.

Structures Obtained:

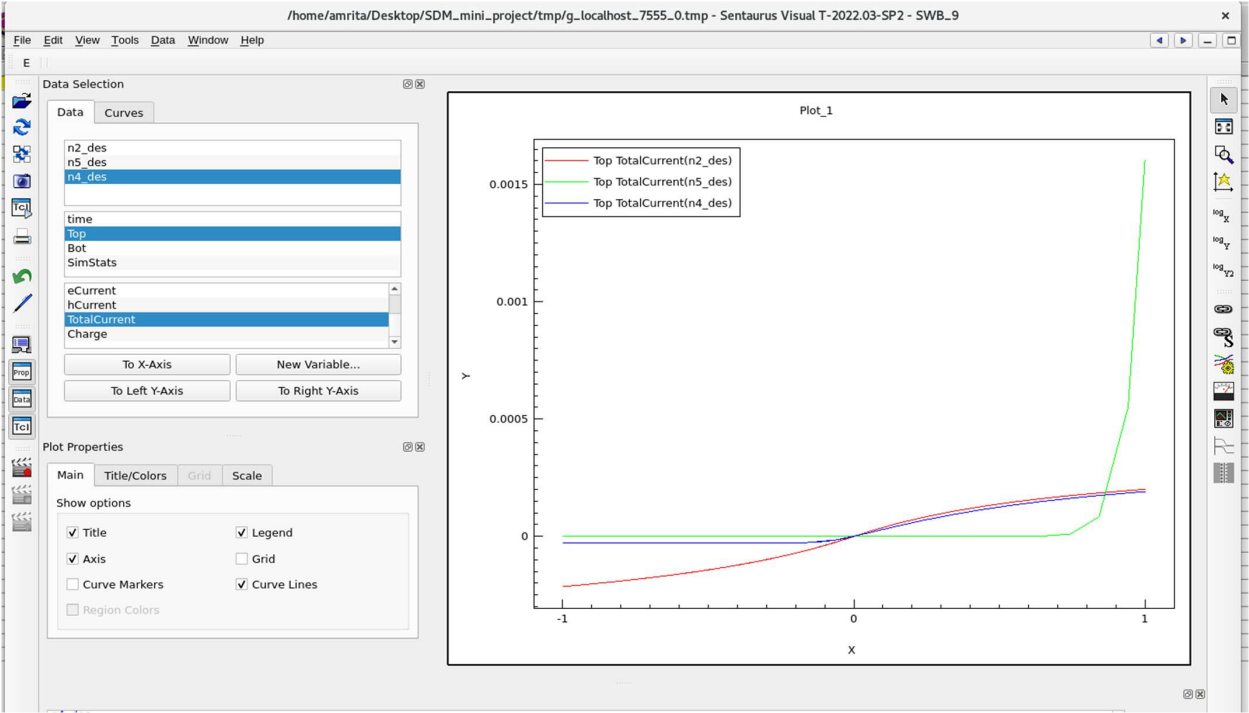
1. Generated Silicon Semiconductor Structure:



2. Doping Concentration:



3. Obtained I – V Curve:



Explanation:

The graph shows the I-V characteristics of four different semiconductor PN junction diodes: Germanium (Ge), Silicon (Si), Gallium Arsenide (GaAs), and 4H-Silicon Carbide (4H-SiC). The voltage is varied from reverse bias to forward bias (+1.5 V), and the current produced by the device is observed.

Red curve → Germanium (Ge), bandgap = 0.66 eV

Blue curve → Silicon (Si), bandgap = 1.12 eV

Green curve → Gallium Arsenide (GaAs), bandgap = 1.42 eV

Cyan curve → 4H-Silicon Carbide (4H-SiC), bandgap = 3.26 eV

What is p-type Silicon?

The Silicon used in this project is p-type Silicon in the top region because it is doped with Boron. In p-type Silicon, holes are the majority charge carriers. In simple words, holes are the main carriers responsible for carrying current through the Silicon. The bottom region is doped with Phosphorus (n-type), where electrons are majority carriers.

Since all four materials are simulated under the same dimensions and doping concentrations, the main difference between the four simulations is the energy bandgap and material parameters of the semiconductor.

Explanation of the curves :

From the graph, the red Germanium curve shows the lowest turn-on voltage ($V_{on} \approx 0.25$ V) compared with the other three curves. The blue Silicon curve turns on at approximately 0.66 V. The cyan 4H-SiC curve turns on at 0.67 V, while the green Gallium Arsenide curve shows a higher turn-on voltage of 0.81 V.

This difference occurs because the four materials have different energy bandgaps. In our simulation:

Germanium → 0.66 eV

Silicon → 1.12 eV

Gallium Arsenide → 1.42 eV

4H-Silicon Carbide → 3.26 eV

For semiconductor diodes, increasing the energy bandgap increases the built-in potential barrier (V_{bi}). Therefore, Germanium turns on at a much lower voltage than Silicon, GaAs, and 4H-SiC.

Why do Silicon and Germanium show different Turn-on Voltages?

The built-in potential barrier $V_{bi} = V_t \cdot \ln(N_A N_D / n_i^2)$ directly governs the forward turn-on voltage. Because Germanium has a much smaller bandgap (0.66 eV), its intrinsic carrier concentration n_i is roughly 2,200 times higher than that of Silicon at 300 K. This substantially lowers the equilibrium barrier to 0.37 V in Germanium, compared to 0.75 V in Silicon. Consequently, majority carrier injection begins at just 0.25 V in Germanium.

Note: In the reverse bias regime, Germanium exhibits severe thermal generation leakage (0.295 nA) due to its high n_i , while Silicon provides near-ideal reverse dark leakage suppression (0.14 fA).

Why is Gallium Arsenide & 4H-SiC different?

Gallium Arsenide has a direct bandgap (1.42 eV), meaning electrons in the conduction band minimum can transition directly to the valence band without requiring momentum exchange with lattice phonons. Under forward bias, injected carriers undergo spontaneous radiative recombination, emitting photons at $\lambda = 873$ nm and functioning as an efficient light-emitting diode (LED).

4H-Silicon Carbide has an ultra-wide bandgap (3.26 eV) and an exceptionally large critical breakdown field (2.2 MV/cm, roughly tenfold higher than Silicon). Consequently, 4H-SiC diodes sustain reverse voltages exceeding 1200 V without entering avalanche breakdown, making it the premier material for high-power electric vehicle inverters.

Conclusion:

The semiconductor PN junction diode was successfully simulated across Germanium, Silicon, Gallium Arsenide, and 4H-Silicon Carbide on the same structure in Synopsys Sentaurus TCAD. The I–V curves show that changing the semiconductor material fundamentally affects the current, turn-on voltage, and overall behaviour of the diode. Germanium shows early turn-on at 0.25 V with high leakage, Silicon delivers balanced switching and sub-femtoamp leakage, Gallium Arsenide provides efficient optoelectronic photon emission, and 4H-Silicon Carbide produces extreme high-voltage breakdown capability. Thus, the simulation demonstrates that the choice of semiconductor material plays a decisive role in the electrical characteristics of a PN junction diode.

Group Members :

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Signature of the staff